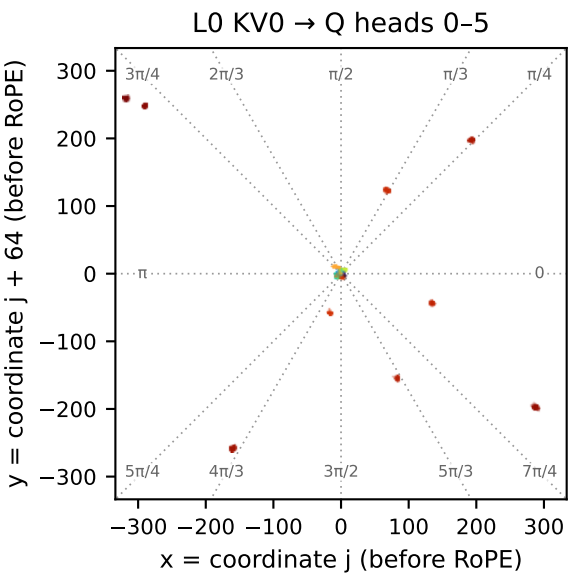
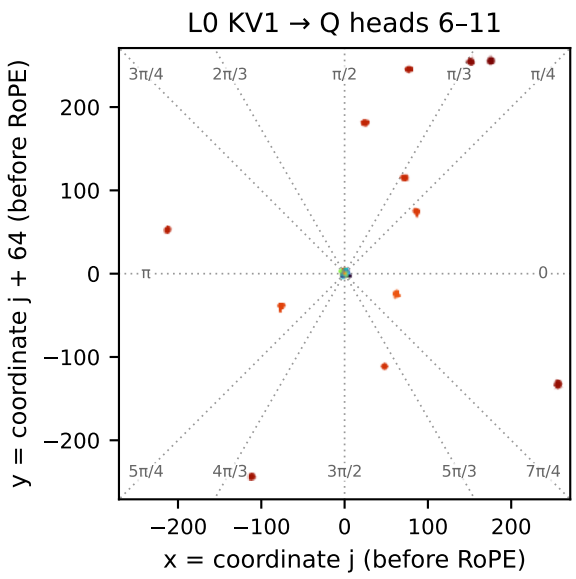
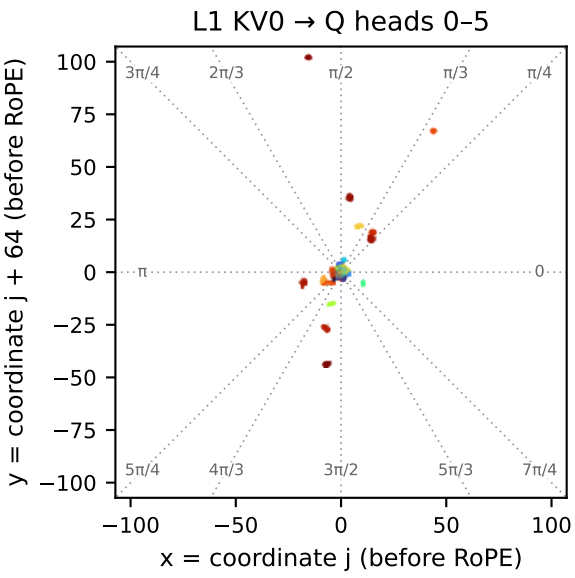
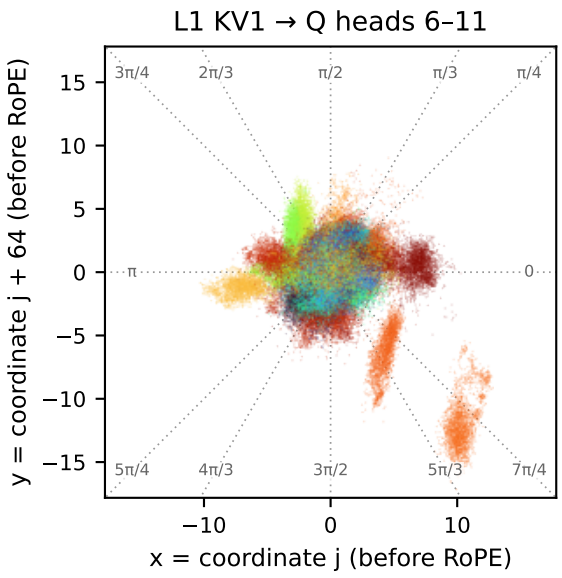
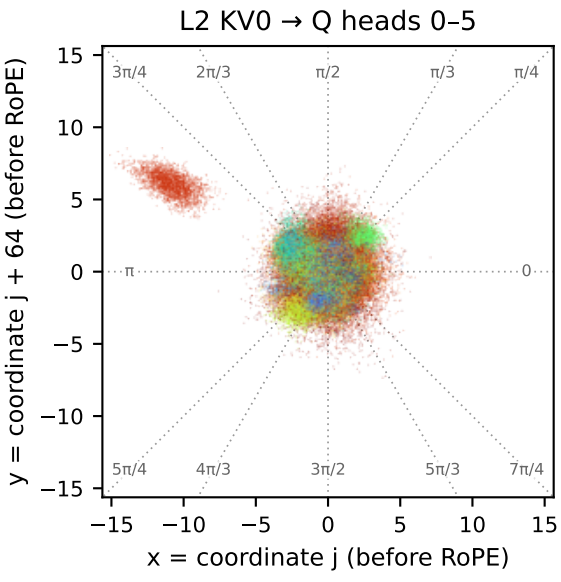
Qwen keys · layer 0–3 · axes fitted separately for each head

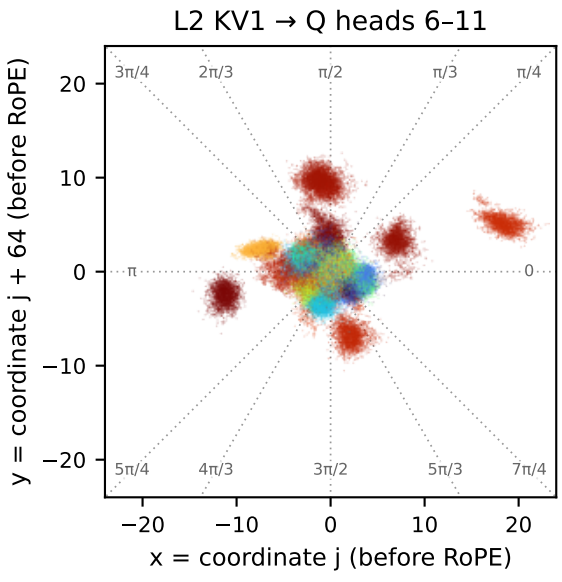


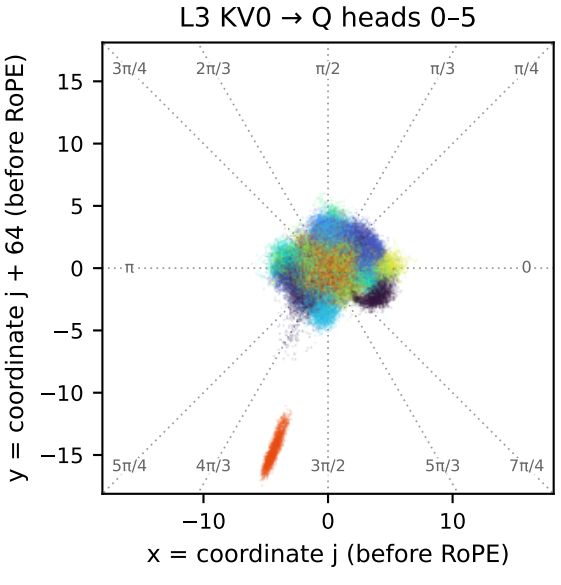


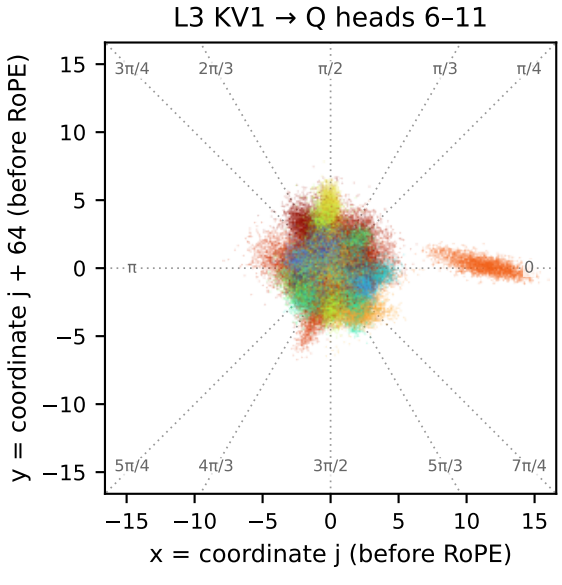


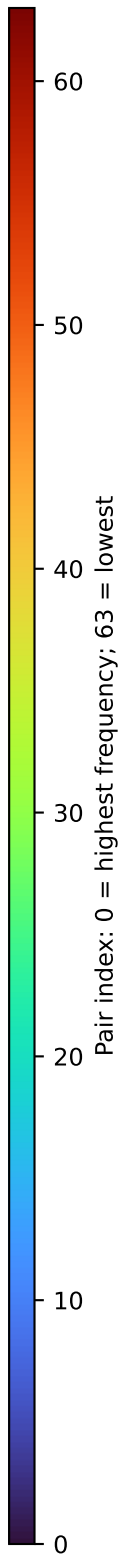






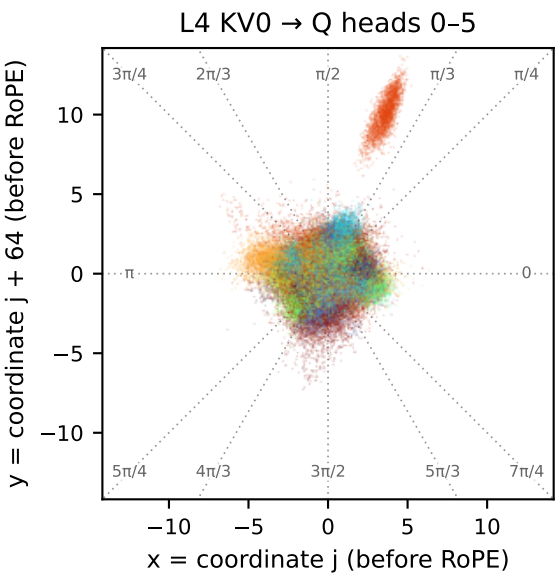


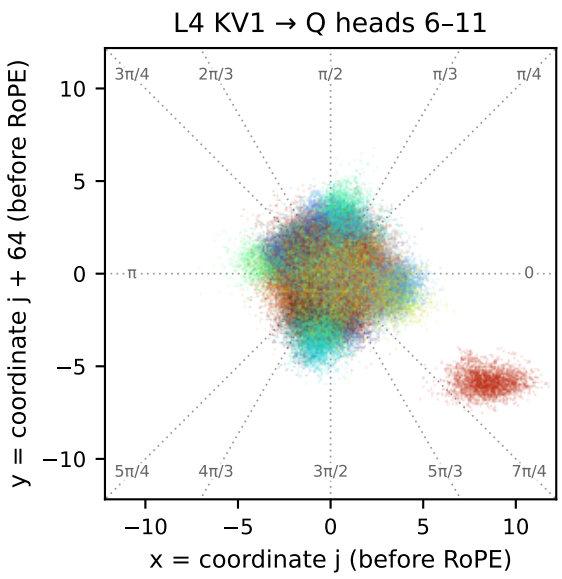


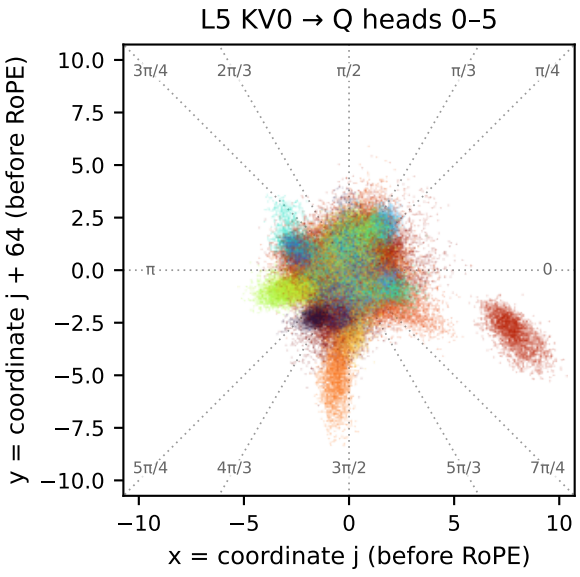


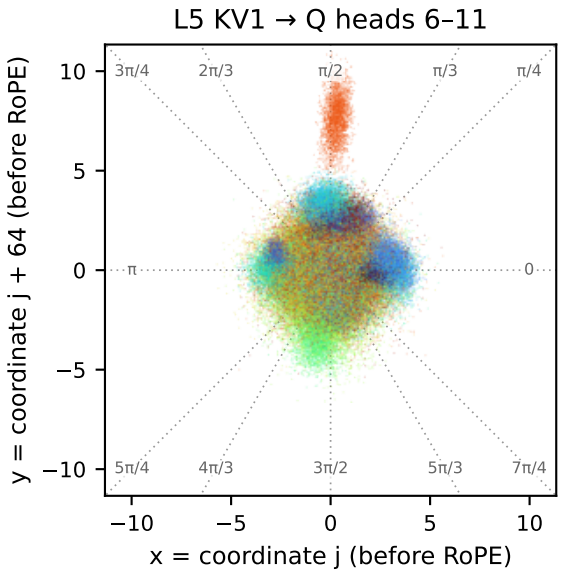
Each point = one token × one RoPE pair. 2,379 prose key tokens; chat-prefix keys excluded; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

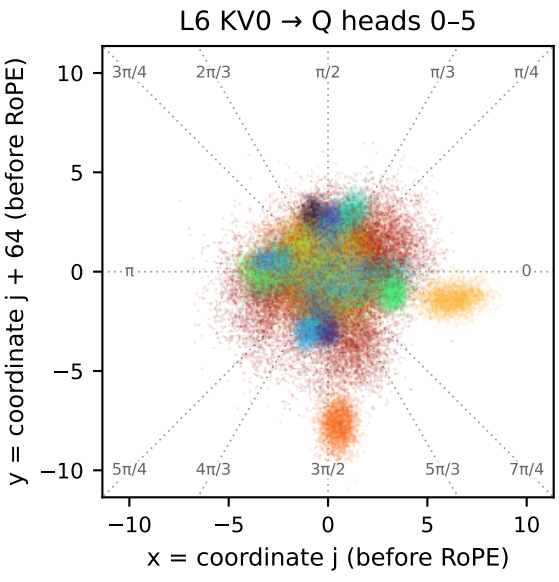
Qwen keys · layer 4–7 · axes fitted separately for each head

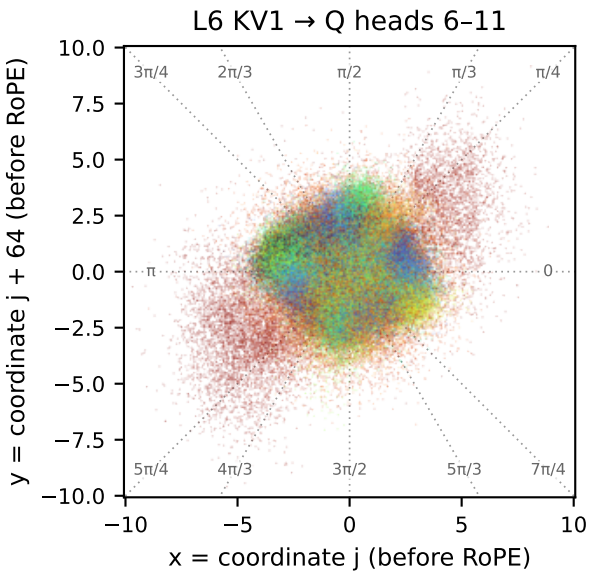


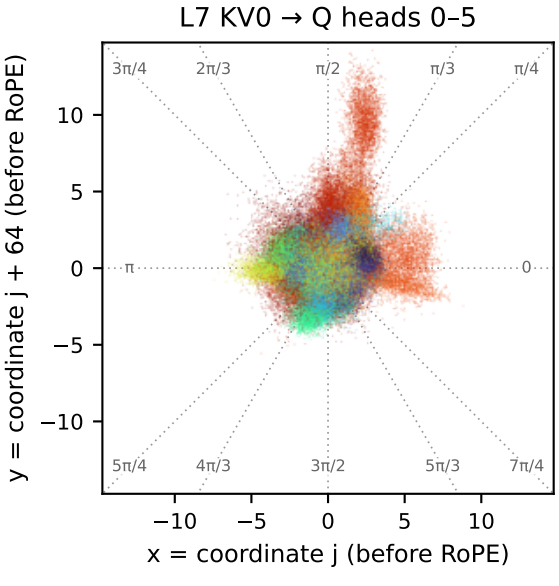


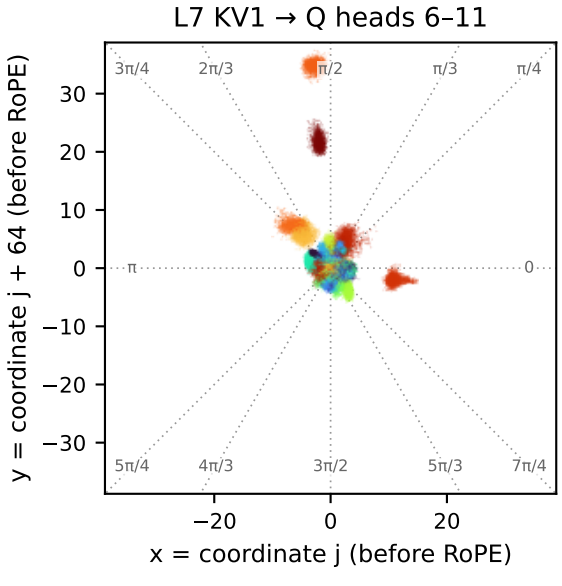


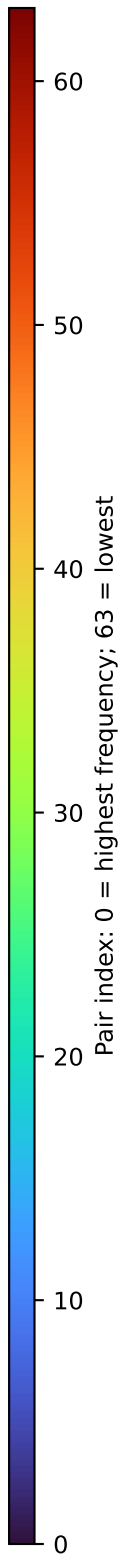






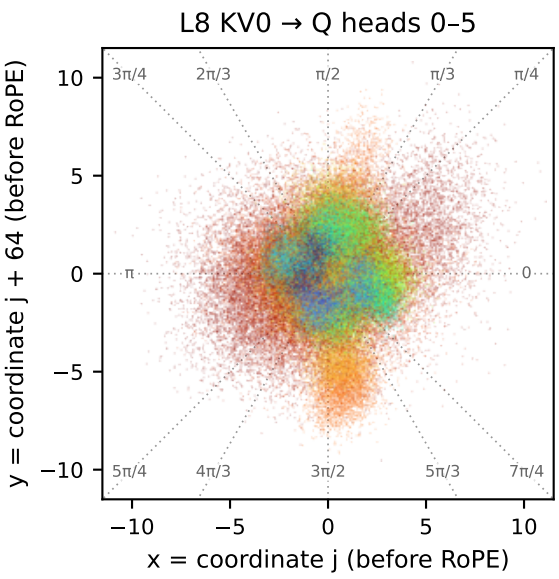


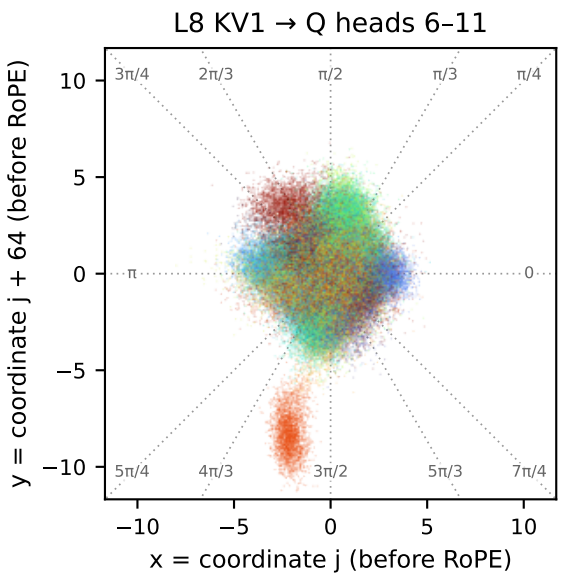


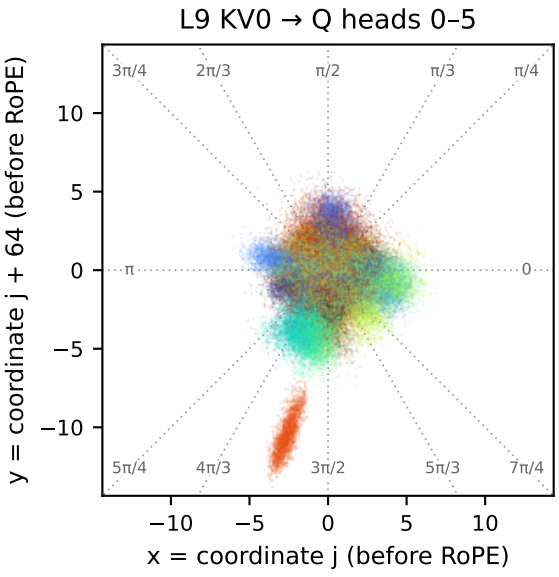


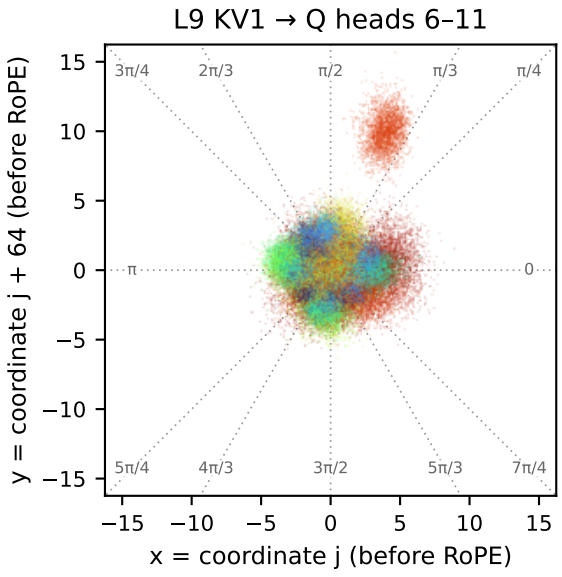
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Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

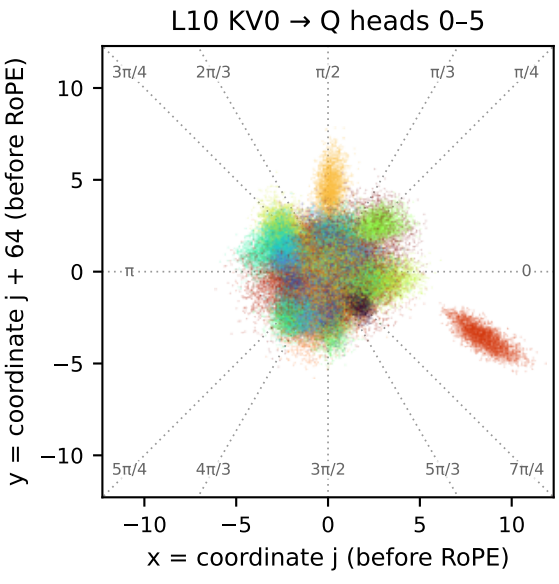
Qwen keys · layer 8–11 · axes fitted separately for each head

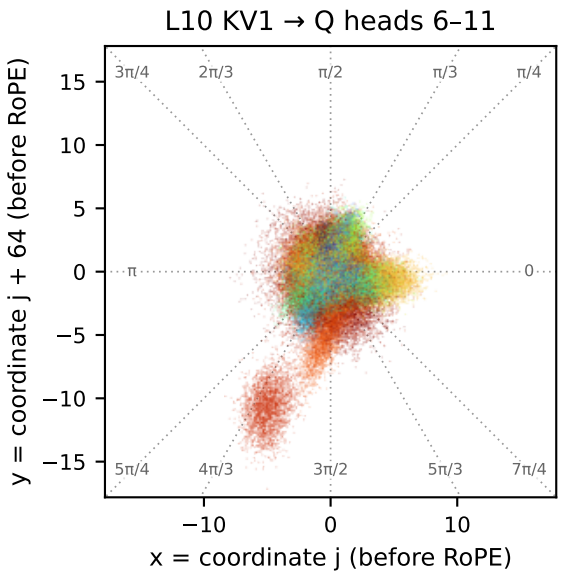


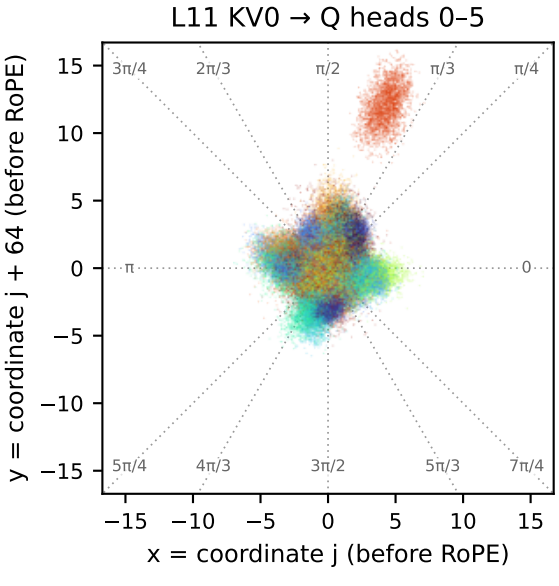


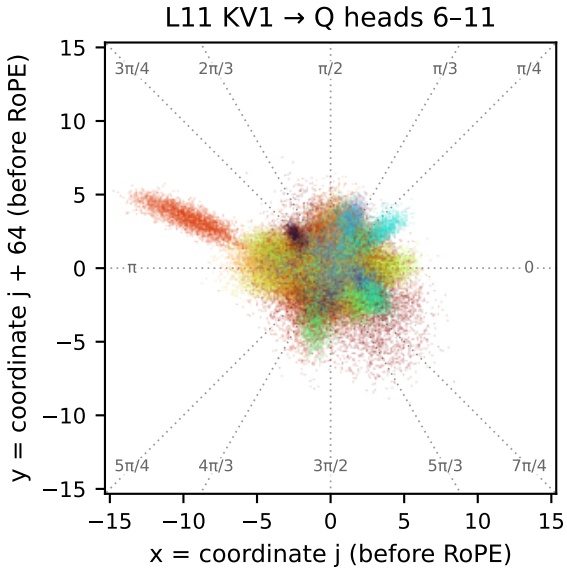


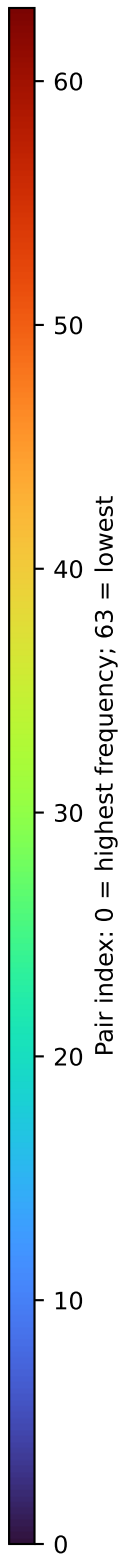






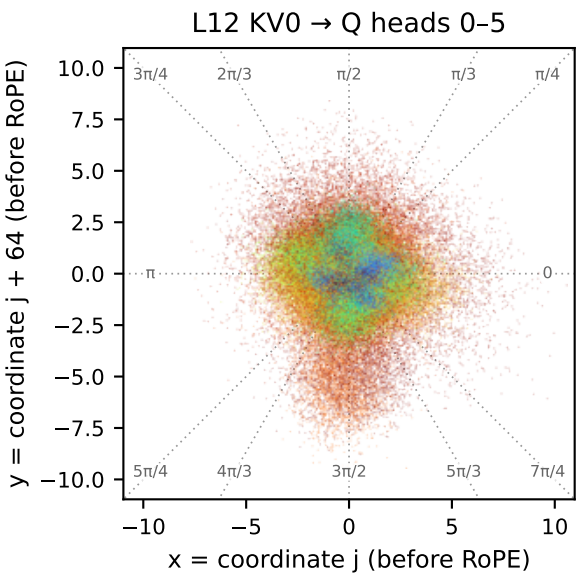


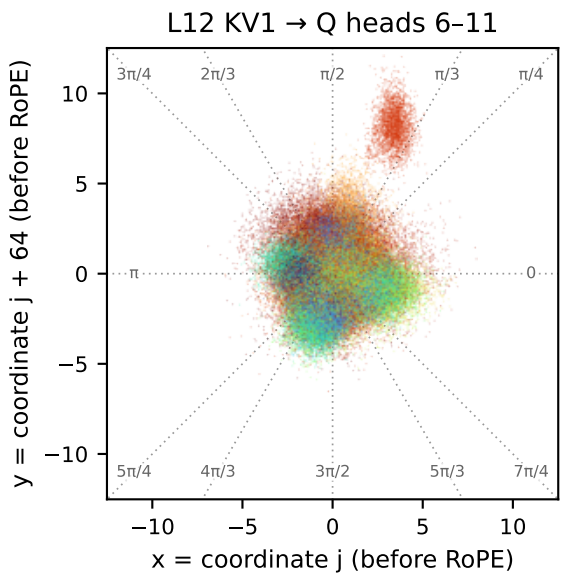


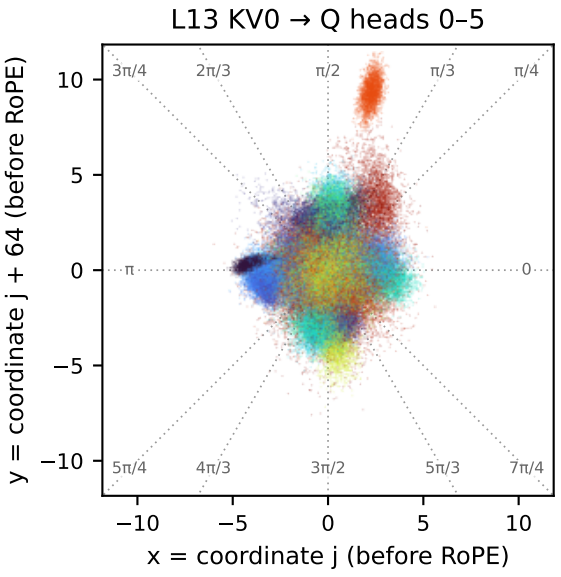


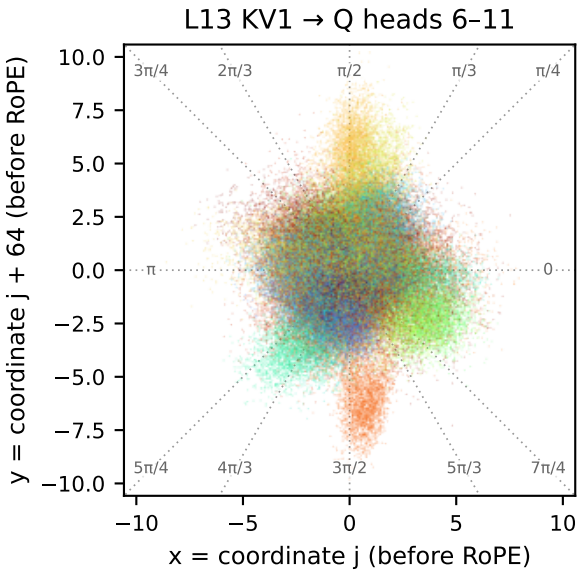
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Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

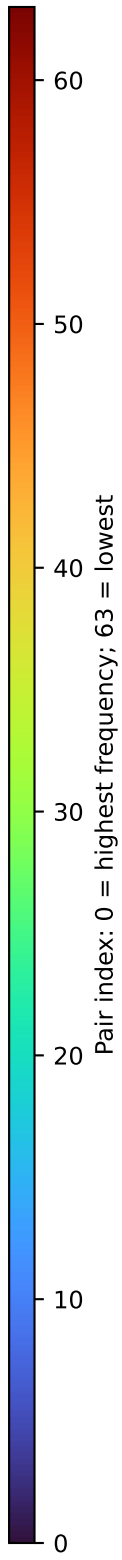
Qwen keys · layer 12–15 · axes fitted separately for each head

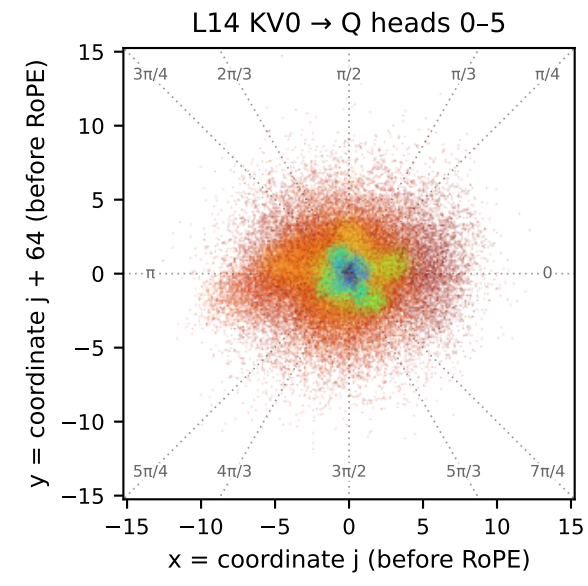


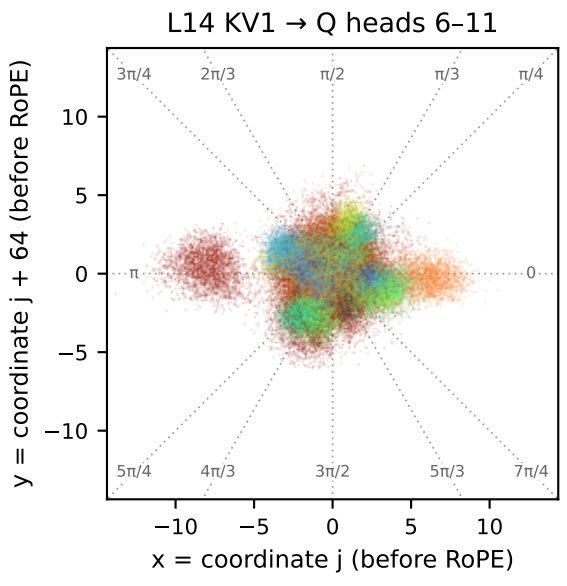


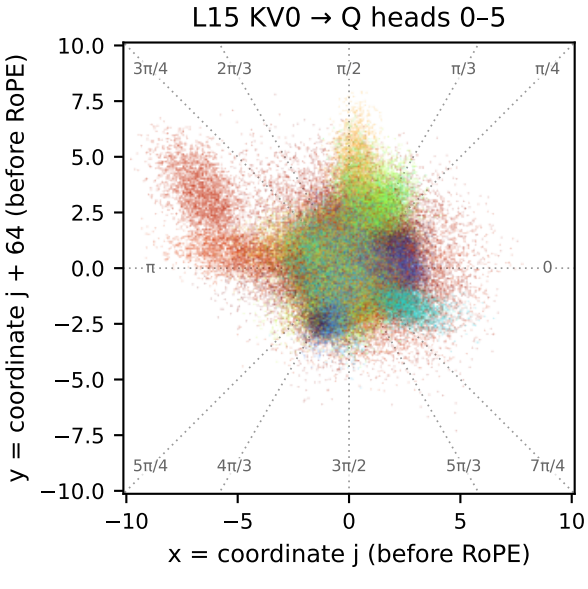


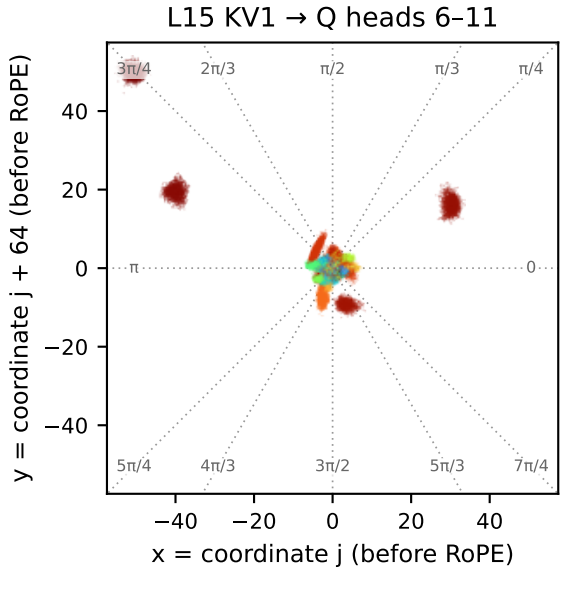






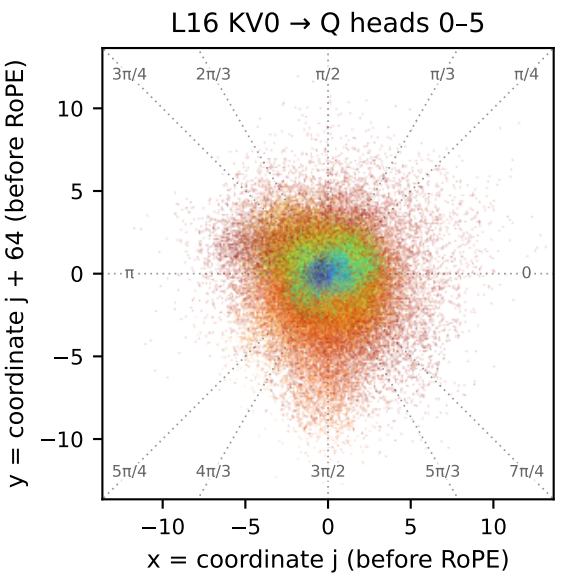


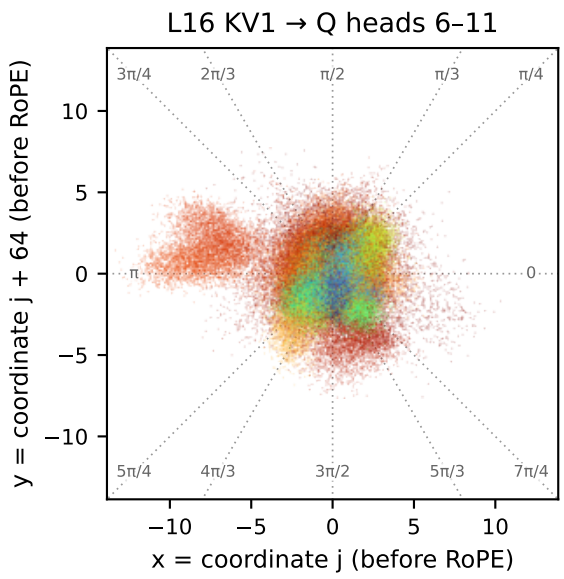


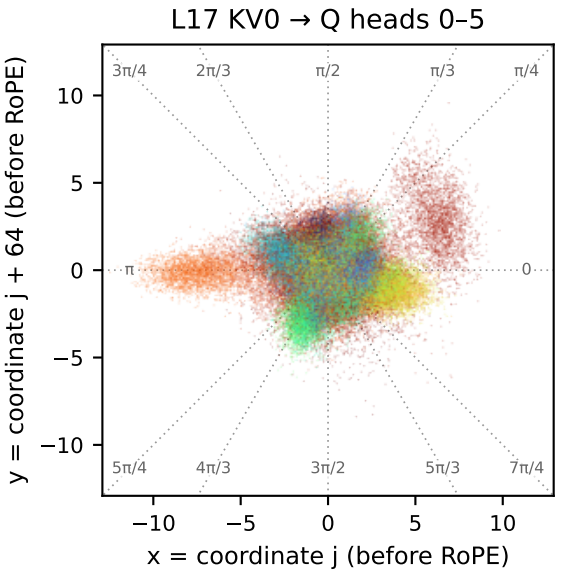


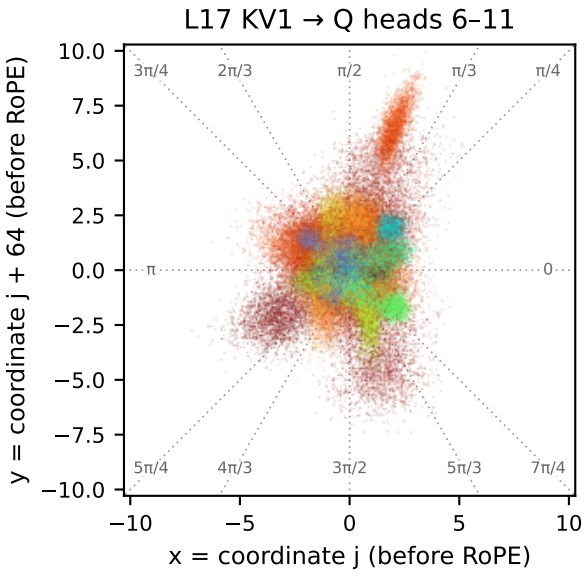
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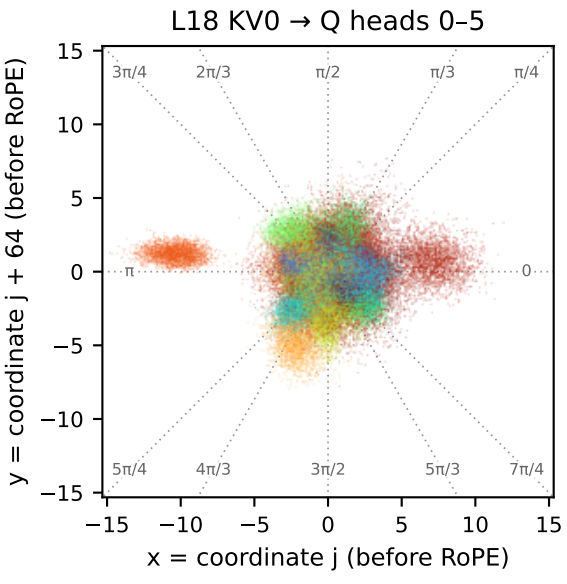
Qwen keys · layer 16–19 · axes fitted separately for each head

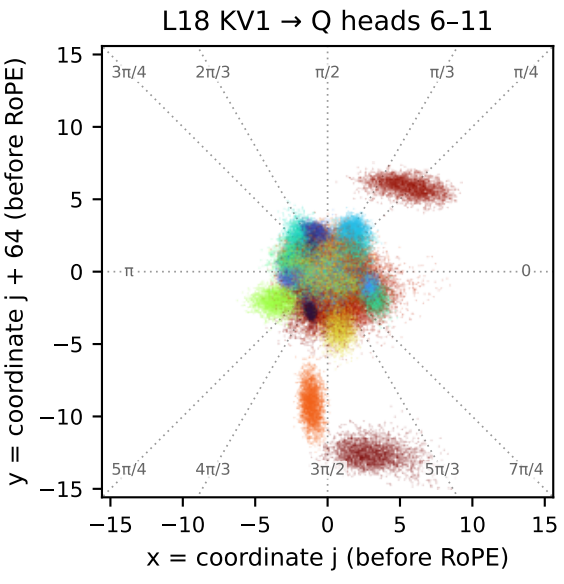


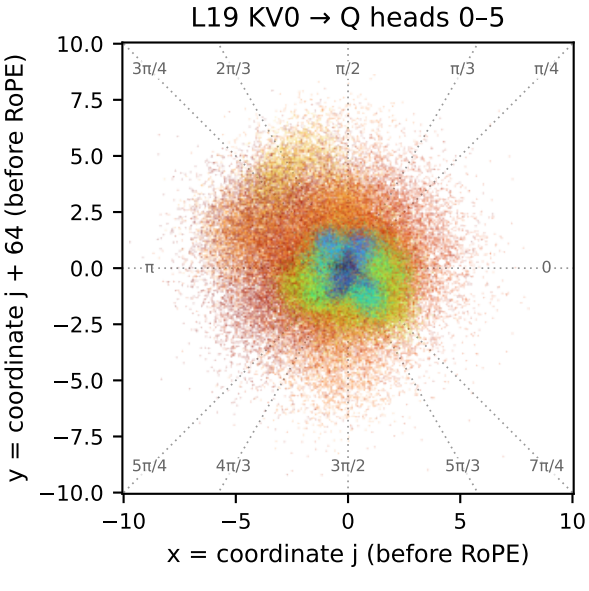


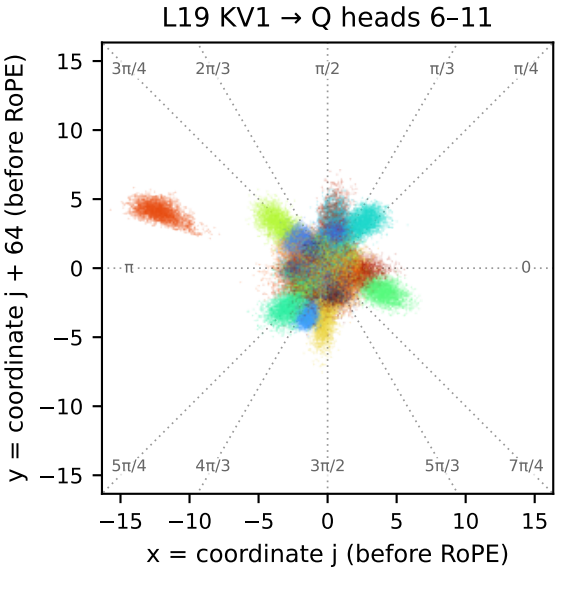


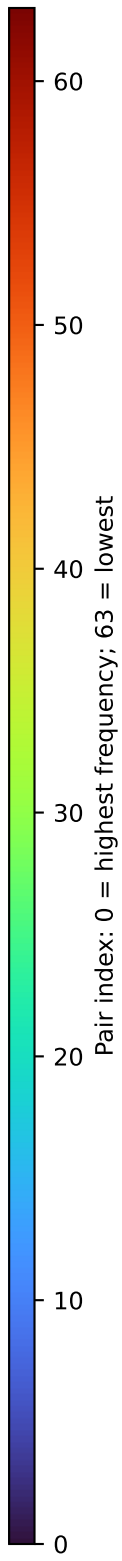






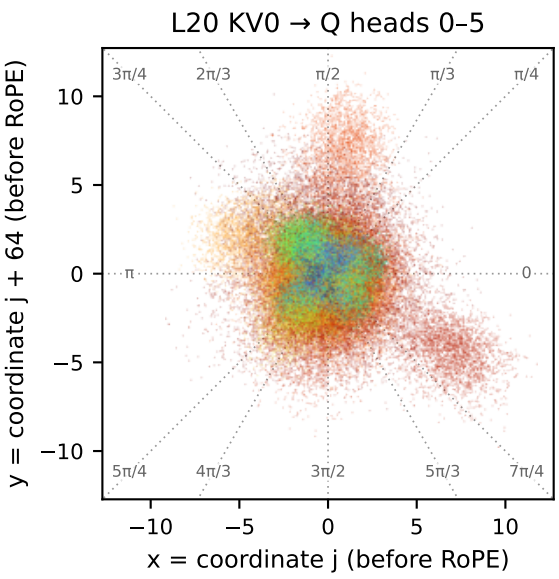


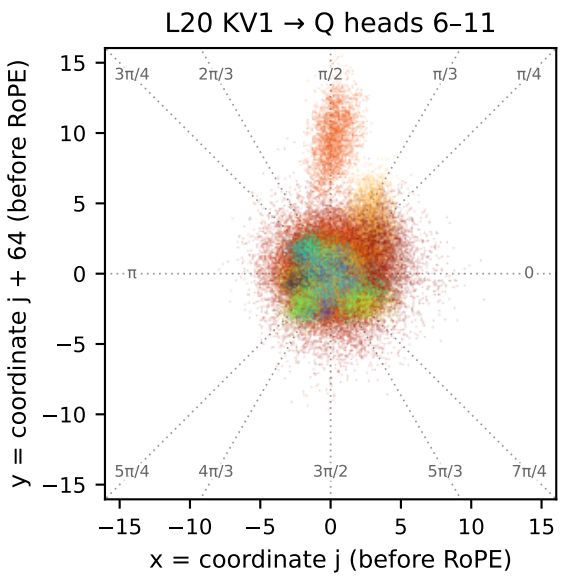


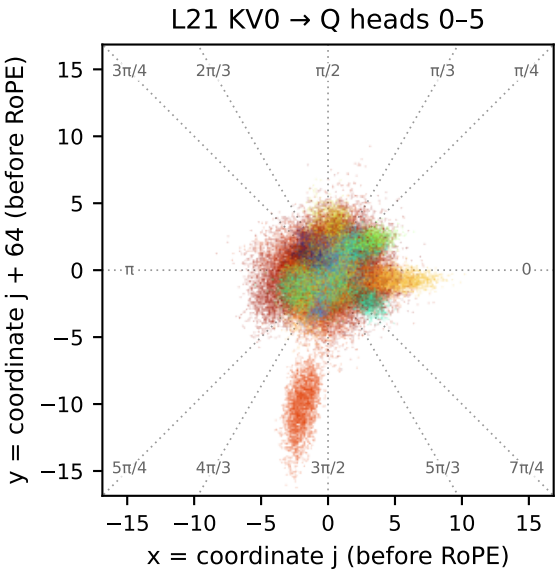


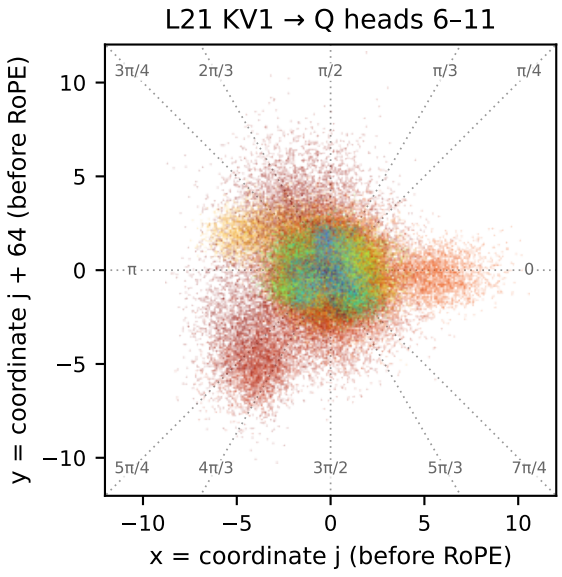
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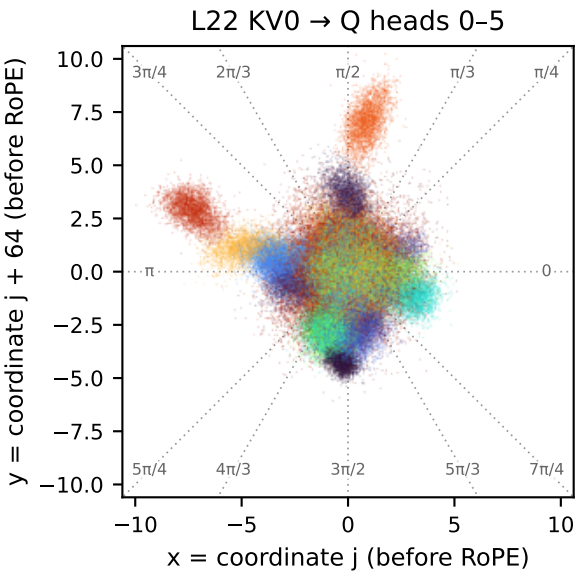
Qwen keys · layer 20–23 · axes fitted separately for each head

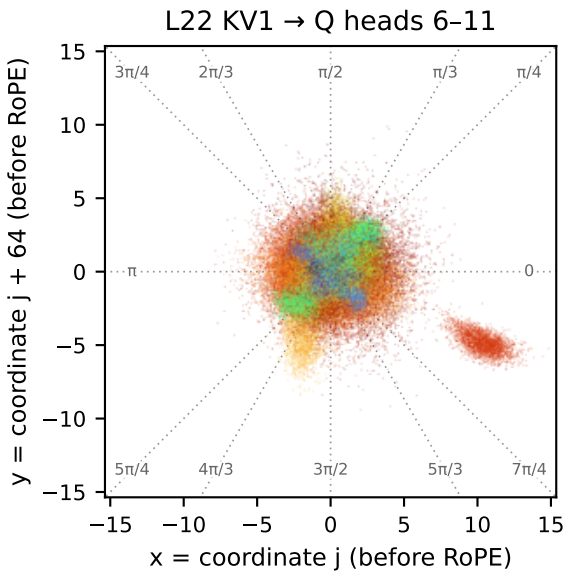


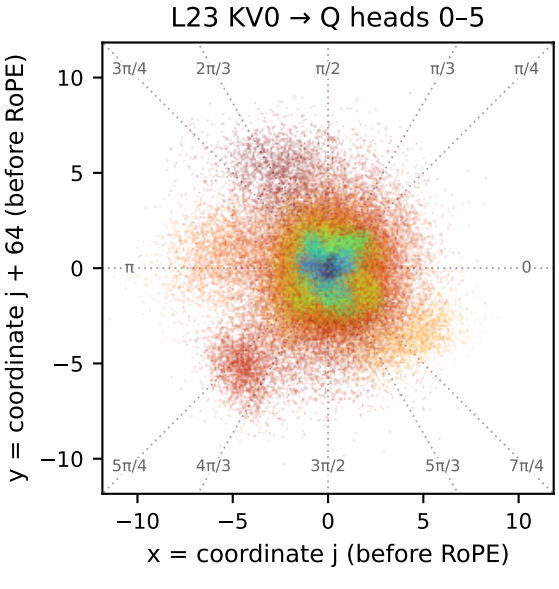


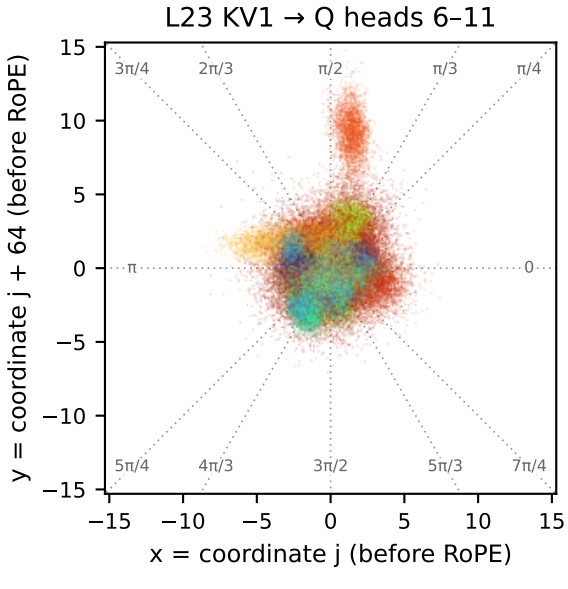


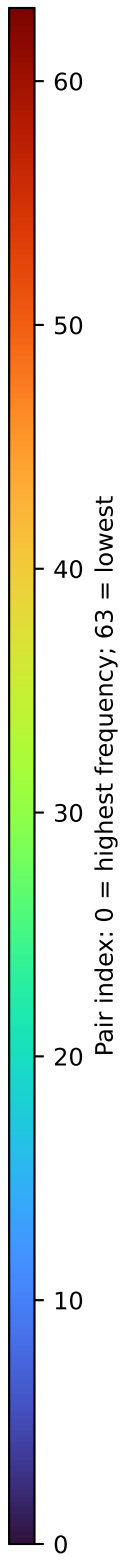






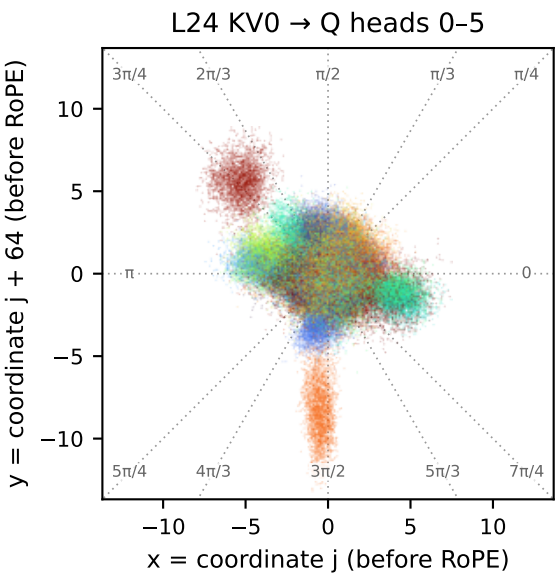


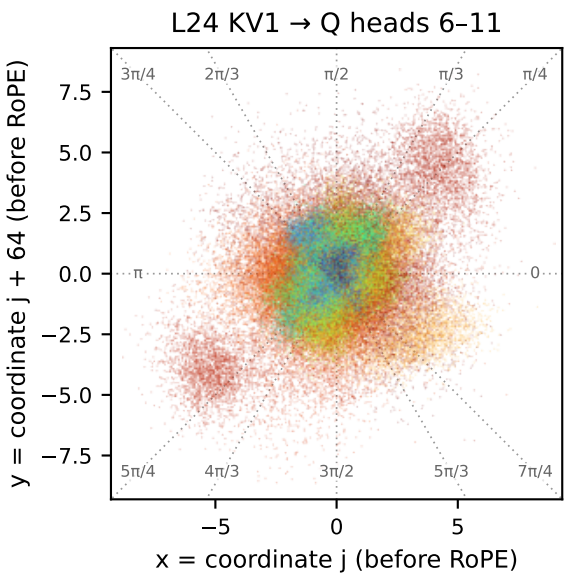


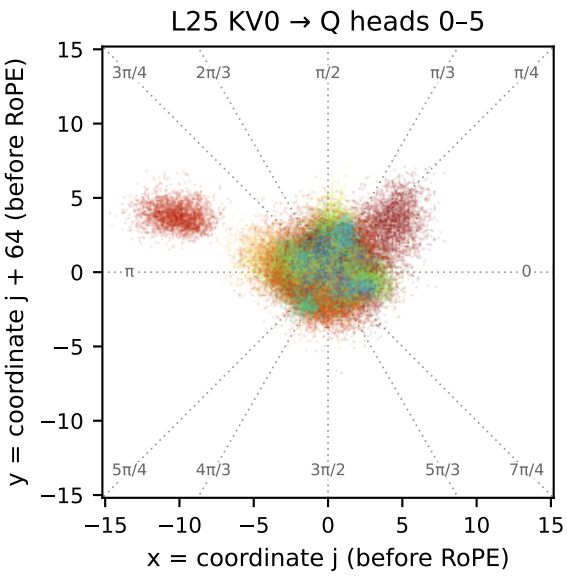


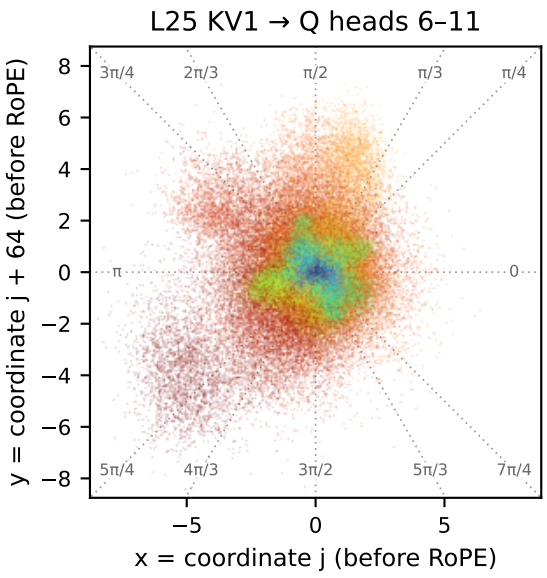
Each point = one token × one RoPE pair. 2,379 prose key tokens; chat-prefix keys excluded; 64 pairs per head.
Known positional rotation removed. Linear axes; no clipping; full range fitted per head.
Within each panel, x and y have the same scale. Axis limits differ between panels; compare tick values.
Gray dotted guides mark multiples of π/4 and π/3.

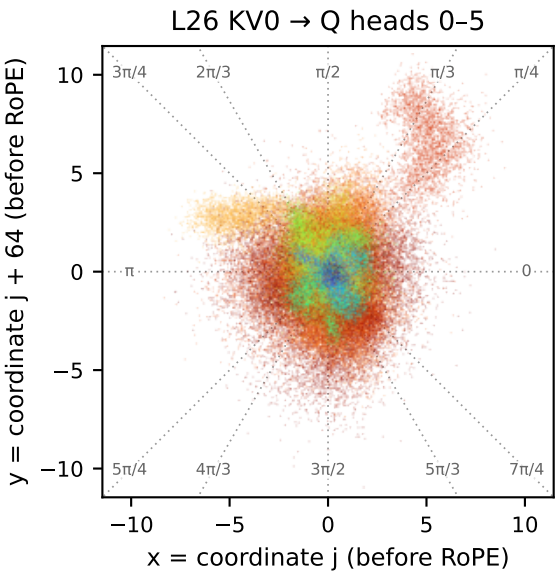
Qwen keys · layer 24–27 · axes fitted separately for each head

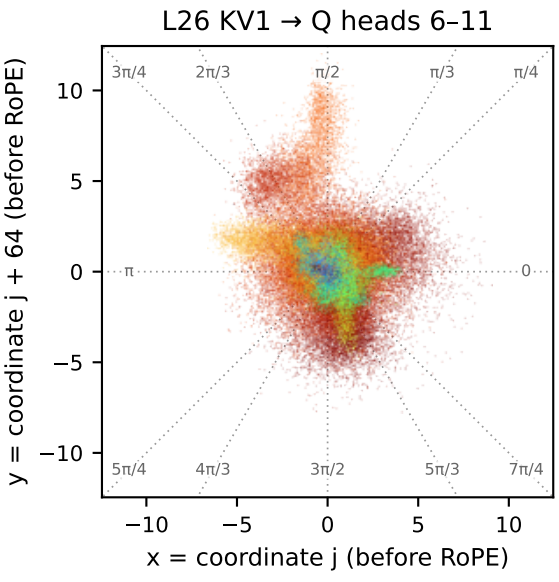


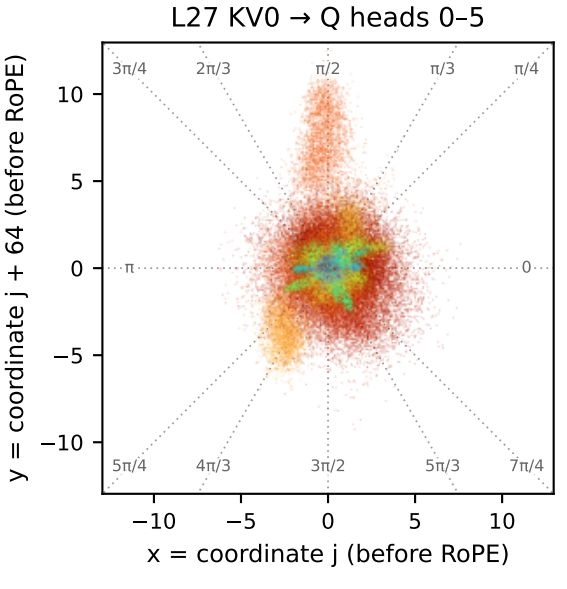


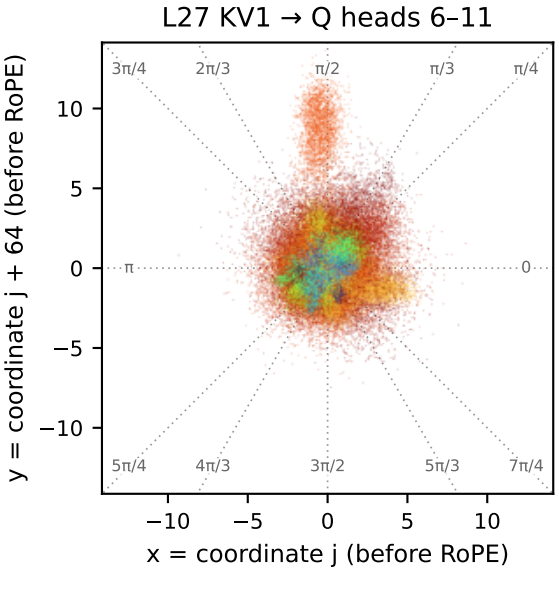


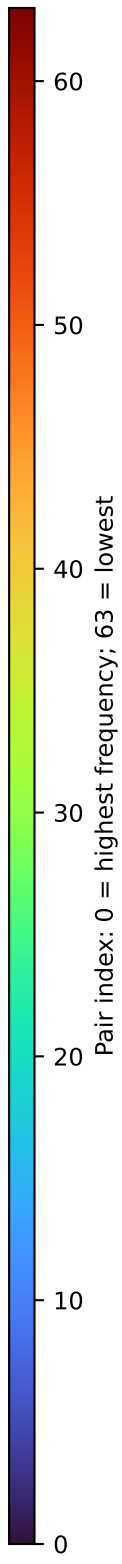












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Gray dotted guides mark multiples of π/4 and π/3.